

SOHAM GHOSH

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RESEARCH INTERESTS

- **Algebraic geometry with an arithmetic flavour:** Brauer groups, Azumaya algebras, twisted sheaves, moduli and stacks, K-theory.
- **Tropical and non-archimedean geometry:** intersection theory of non-archimedean cycles and currents, tropical cohomology, tropical abelian varieties, and Fourier–Mukai transforms.
- **Birational geometry:** study of various positivity conditions on bundles, foliations and resulting characterizations of special Fano varieties.
- **Commutative algebraic aspects of algebraic geometry**

EDUCATION

University of Washington, Seattle, WA <i>Ph.D. in Mathematics</i> Advisors: Max Lieblich and Farbod Shokrieh	2022–Present
University of Washington, Seattle, WA <i>M.S. in Mathematics</i> GPA: 4.0/4.0	2022–2024
Indian Statistical Institute, Bangalore, India <i>Bachelor of Mathematics (Honors)</i> Grade: 97.18%, First division with distinction	2019–2022

PUBLICATIONS AND PREPRINTS

- [1] *Characterizations of products of projective spaces and hyperquadrics* (with Sándor Kovács), in preparation.
- [2] *Torelli theorems for moduli stacks and twisted moduli spaces of vector bundles* (with Ting Gong and Max Lieblich), preprint (2026). [arXiv:2609.29011](#)
- [3] *Characterizing contractions and weighted blowdowns* (with Tyson Klingner), preprint (2026). [arXiv:2609.04023](#)
- [4] *Kovács’ conjecture on characterization of projective space and hyperquadrics*, preprint (2026). [arXiv:2601.10055](#)
- [5] *Tropical Poincaré bundle, Fourier–Mukai transform, and a generalized Poincaré formula* (with Farbod Shokrieh), preprint (2025). [arXiv:2503.12835](#)
- [6] *The Zariski covering number for vector spaces and modules*, *Communications in Algebra* **50** (2022), No. 5, pages 1994–2017. [doi:10.1080/00927872.2021.1995741](#)

SELECTED AWARDS AND HONORS

Lee and Cozette McFarlan Travel Fellowship, University of Washington For travel to attend conferences and give talks	2026
Jane Street Graduate Research Fellowship Finalist, among applicants across EECS, mathematics, physics, statistics	2026

Simons Dissertation Fellowship in Mathematics 40 fellowships awarded annually across US, by the Simons Foundation	2025
Selected for Summer Research Institute in Algebraic Geometry 2025 Bootcamp Fully funded	Summer 2025
Departmental Research Assistantship, University of Washington, Seattle	Summer 2025
Graduate School Conference Presentation Award, University of Washington, Seattle	2024
Graduate Research Assistantship, University of Washington, Seattle Partially funded by NSF CAREER DMS-2044564 and NSF FRG DMS-2151718 grants	Summer 2024
Academic Excellence Award, University of Washington, Seattle	2023
Provost Scholarship, University of Washington, Seattle	2022 & 2023
Academic Excellence Award, Indian Statistical Institute, Bangalore	2020, 2021, & 2022

INVITED TALKS

Kennesaw, Georgia, AMS Fall Southeastern Sectional Meeting Special session on “Geometry and topology of hypersurfaces”	October 2026
Indian Statistical Institute (ISI) Kolkata, Department Colloquium	September 2026
Junior Algebraic Geometry Workshop 2026, University of California Los Angeles (UCLA)	July 2026
University of California San Diego (UCSD), Algebraic Geometry Seminar	May 2026
Indian Institute of Science (IISc) Bangalore, Geometry and Topology Seminar (joint with Algebra and Combinatorics Seminar)	August 2025
Indian Statistical Institute (ISI) Bangalore, Department Colloquium	August 2025
University of Washington (UW), Algebra and Algebraic Geometry Seminar	May 2025
University of Washington (UW), Student Algebraic Geometry Seminar	May 2025, February 2026
University of California Los Angeles (UCLA), Birational Geometry Seminar	May 2025
San Antonio, Texas, AMS Fall Central Sectional Meeting Special session on “Non-Archimedean, Algebraic, tropical geometry and Applications”	September 2024

CONFERENCES AND WORKSHOPS

• AMS Fall Southeastern Sectional Meeting at Kennesaw State University, Georgia	October 2026
• Junior Algebraic Geometry Workshop at University of California Los Angeles	July 2026
• Western Algebraic Geometry Symposium Spring 2026 at University of California San Diego	March 2026
• Summer Research Institute in Algebraic Geometry 2025 Weeks 1–3 at Colorado State University	July 2025
• Summer Research Institute in Algebraic Geometry 2025 Bootcamp at Colorado State University	July 2025
• Arithmetic, K-theory and Algebraic cycles 2025 at Ohio State University	May 2025
• Western Algebraic Geometry Symposium Fall 2024 at University of Arizona, Tucson	November 2024
• AMS Fall Central Sectional Meeting at University of Texas San Antonio	September 2024
• Arizona Winter School 2024: Abelian Varieties	March 2024
• Western Algebraic Geometry Symposium Fall 2023 at Washington University St. Louis	November 2023
• FRG workshop on Brauer groups and derived categories at Northwestern University	October 2023
• Western Algebraic Geometry Symposium Spring 2023 at the University of Washington	April 2023
• Arizona Winter School 2023: Unlikely intersections	March 2023

TEACHING

Teaching Assistant and Grader, *University of Washington, Seattle*

Autumn 2022–Present

Math 124	Calculus I	Autumn 2022, 2 sections
Math 125	Calculus II	Winter 2023, 2 sections
Math 327	Introduction to Real Analysis I	Spring 2023
Math 394	Probability I	Summer 2023
Math 381	Discrete Mathematical Modelling	Autumn 2023
Math 407	Linear Optimization	Winter 2024, 2 sections
Math 534	Graduate Complex Analysis	Spring 2024 and Autumn 2024
Math 300	Mathematical Reasoning	Winter 2025
Math 424	Fundamental Concepts of Analysis	Spring 2025
Math 402	Introduction to Modern Algebra: Rings and Fields	Autumn 2025
Math 505	Graduate Algebra: Galois theory	Winter 2026
Math 404	Introduction to Modern Algebra: Galois theory	Spring 2026
Math 504	Graduate Algebra: Group theory	Autumn 2026

MENTORING AND OUTREACH

Mentor, Washington Directed Reading Program for undergraduates

2026–

University of Washington, Seattle

Mentored Larry Wen on Commutative Algebra following Atiyah–Macdonald, and Max Han on Representation theory following Serre’s *Linear Representations of Finite Groups* during Spring 2026.

Instructor, Husky Math Academy

2024–2025

University of Washington, Seattle

Instructor for weekly problem-solving and discussion sessions for grades 7–9 students, covering diverse and advanced topics and designing classroom activities. Link: <https://www.huskymath.org>.

SERVICE

Refereeing for journals

Communications in Algebra, Journal of Commutative Algebra, Journal of Algebraic Combinatorics, American Mathematical Monthly, Indian Journal of Pure and Applied Mathematics.

OTHER ACADEMIC ACHIEVEMENTS

- **Indian Institute of Technology** Joint Entrance Exam (IIT-JEE) Advanced 2019: **All India Rank (AIR) 878** out of 161,319 students shortlisted from JEE-Mains
- **Indian Institute of Technology** Joint Entrance Exam (IIT-JEE) Mains 2019: **All India Rank (AIR) 1516** out of 1,147,125 students
- **Indian Statistical Institute** Entrance Exam 2019: **All India Rank (AIR) 27** out of $\approx 50,000$ students
- **Chennai Mathematical Institute** Entrance Exam 2019: offered *Sriram fellowship*, covering full tuition fee and a monthly stipend for the course duration, awarded to top scorers in the exam (*declined*)
- Selected for **Indian National Mathematics Olympiad 2018** via Regional Mathematics Olympiad 2017.
- **Kishore Vaigyanik Protsahan Yojana (KVPY)** Exam 2017: **All India Rank (AIR) 201** out of $\approx 60,000$ students. Awarded fellowship for bachelor’s and master’s studies in sciences in India, funded by the Department of Science and Technology, Government of India.

REFERENCES

Research

Max Lieblich

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Teaching

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